

Question description

You are given an array A of N integers. Now, two functions $F(X)$ and $G(X)$ are defined:

$F(X)$: This is the smallest number Z such that $X < Z \leq N$ and $A[X] < A[Z]$

$G(X)$: This is the smallest number Z such that $X < Z \leq N$ and $A[X] > A[Z]$

Now, you need to find for each index i of this array $G(F(i))$, where $1 \leq i \leq N$. If such a number does not exist, for particular index i , output 1 as its answer. If such a number does exist, output $A[G(F(i))]$

Constraints:

$$1 \leq N \leq 3000$$

$$0 \leq A[i] \leq 10^{18}$$

Input :

The first line contains a single integer N denoting the size of array A . Each of the next N lines contains a single integer, where the integer on the i^{th} line denotes $A[i]$.

Output :

Print N space-separated integers on a single line, where the i^{th} integer denotes $A[G(F(i))]$ or 1 , if $G(F(i))$ does not exist.

▼ Logical Test Cases

Test Case 1

INPUT (STDIN)

8
3 7 1 7 8 4 5 2

EXPECTED OUTPUT

1 4 4 4 -1 2 -1 -1

Test Case 2

INPUT (STDIN)

8
31 71 11 17 81 14 51 21

EXPECTED OUTPUT

11 14 14 14 -1 21 -1 -1

▼ Mandatory Test Cases

Test Case 1

KEYWORD

if(a[j]>a[i])

Test Case 2

KEYWORD

for(i=0;i<n;i++)

▼ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

9

Test Case 2

TOKEN COUNT

225

Test Case 3

NLOC

30

```
Stack7.c +
1 #include <stdio.h>
2
3 int main() {
4     int n;
5     if (scanf("%d", &n) != 1) return 0;
6     long long a[n];
7     for (int i = 0; i < n; i++) {
8         scanf("%lld", &a[i]);
9     }
10
11     int next_greater[n];
12     for (int i = 0; i < n; i++) {
13         next_greater[i] = -1;
14         for (int j = i + 1; j < n; j++) {
15             if (a[j] > a[i]) {
16                 next_greater[i] = j;
17                 break;
18             }
19         }
20     }
21
22     int next_smaller[n];
23     for (int i = 0; i < n; i++) {
24         next_smaller[i] = -1;
25         for (int j = i + 1; j < n; j++) {
```

```
26         if (a[j] < a[i]) {
27             next_smaller[i] = j;
28             break;
29         }
30     }
31 }
32
33 int i;
34 for (i = 0; i < n; i++) {
35     int fg = -1;
36     int f = next_greater[i];
```

```
37     if (f != -1) {
38         int g = next_smaller[f];
39         if (g != -1) {
40             fg = a[g];
41         }
42     }
43     printf("%d%c", fg, (i == n - 1) ? '\n' : ' ');
44 }
45 return 0;
46}
47
```

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt

8

3 7 1 7 8 4 5 2

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Stack7.exe

1 4 4 4 -1 2 -1 -1

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt

8

31 71 11 17 81 14 51 21

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Stack7.exe

11 14 14 14 -1 21 -1 -1

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>